

A New AC False Data Injection Attack Method Without Network Information

Runhai Jiao^{ID}, Gangyi Xun, Xuan Liu^{ID}, *Member, IEEE*, and Guangwei Yan

Abstract—Cyberattack emerges as a critical concern in Internet-of-Things integrated power systems. Cybersecurity capabilities of power systems rely on a deep understanding of potential cyberattacks for identifying and protecting against cyber vulnerabilities. The common assumption of system network information in most existing works is suspicious, which fails to hold in practical fields. This paper proposes a novel data-driven false data injection attack method, where only easily accessible measurement data are required, i.e., power injections at buses, flows of lines and the tie line information (connection relationship and power measurement data). In the proposed method, generative adversarial network (GAN) is adopted to extract the physical model using historical measurement data, and a self-attention mechanism is integrated to further capture the power flow laws in the data. After offline training, the effective false data can be constructed in a timely fashion without system network information. The effectiveness of the proposed attack method is validated using the IEEE14 and IEEE118 systems, in which the constructed false data injection attack can evade the system residual detection with an average success rate over 90% under different levels of measuring errors.

Index Terms—False data injection attack, AC state estimation, data-driven, generative adversarial network, self-attention mechanism.

I. INTRODUCTION

WITH the massive adoptions of advanced information and communication technology (ICT), the power system control center usually features sophisticated communication, computation, and control functionalities [1]. Although the networked control system is efficient and convenient, it also incorporates more network intrusion risks, which would deteriorate the security and stable operation of the physical network by tampering with device information or sending wrong instructions [2]–[4]. For instance, many regions of Venezuela, including the capital Caracas, suffered power outages for more than 24 hours in March 2019, which caused a massive traffic jam and public facility collapse, e.g.,

Manuscript received February 23, 2021; revised June 13, 2021; accepted July 19, 2021. Date of publication August 4, 2021; date of current version October 21, 2021. This work was supported by the National Natural Science Foundation of China under Grant 51777062. Paper no. TSG-00307-2021. (*Corresponding author: Xuan Liu.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSG.2021.3102329>.

Digital Object Identifier 10.1109/TSG.2021.3102329

hospitals, schools, etc. Therefore, the cyberattack has emerged as a critical concern for power systems [5].

The substation in the smart grid plays an important role in integrating communication and power infrastructure. Therefore, some works have explored the cyberattack detection and mitigation methods of substation automation to maintain the safe and stable operation of the power grid [6]–[8]. Hong and Liu [9] proposed the new concept of intelligent electronic devices (IEDs) with built-in distributed intrusion detection systems to monitor and detect network attacks. Touhiduzzaman *et al.* [10] applied a game-theoretic graph coloring technique to determine the optimal allocation of security mechanism diversity that minimizes the impact of security vulnerabilities to the grid.

Many works have explored the cyberattack methods [11]–[15]. They analyzed the attack principle and attack conditions in detail, which provides a good reference for the design of subsequent detection and defense strategies. In [16]–[18], false data injection attack (FDIA) is constructed using the linear DC state estimation instead of the nonlinear AC state estimation for facilitating the false data construction. But the constructed false data based on the DC state estimation [19]–[21] is incapable of evading the bad data detection algorithm using the AC state estimation in practical fields [22]. Then, several works adopted AC state estimation to build the FDIA with certain assumptions [23]–[26]. Jia *et al.* [27] assumed that the adversaries can access the power system network information, i.e., network topology and line parameters, to construct the optimal false data by calculating the minimum generalized eigenvalue of the characteristic function. Hug and Giampapa [28] proposed an attack method to tamper with the remote terminal unit measurements based on the estimated power system state. But the assumption of full network information is suspicious, which does not hold in practical fields. Hence, several research efforts have been conducted to analyze FDIA with partial information. After obtaining the network information of the attacking region, Liu and Li [29] approximately calculated the voltage phase angle difference in the boundary buses and then constructed false data by guaranteeing the voltage phase angle difference unchanged, which could effectively avoid the cyberattack detection. Considering the energy distributions and local load redistributions, Fu *et al.* [30] used the minimum residual in the attacking region as the objective function to establish an FDIA model.

In an illustrative SCADA system depicted in Fig. 1 [17], [31], [32], the control center receives the

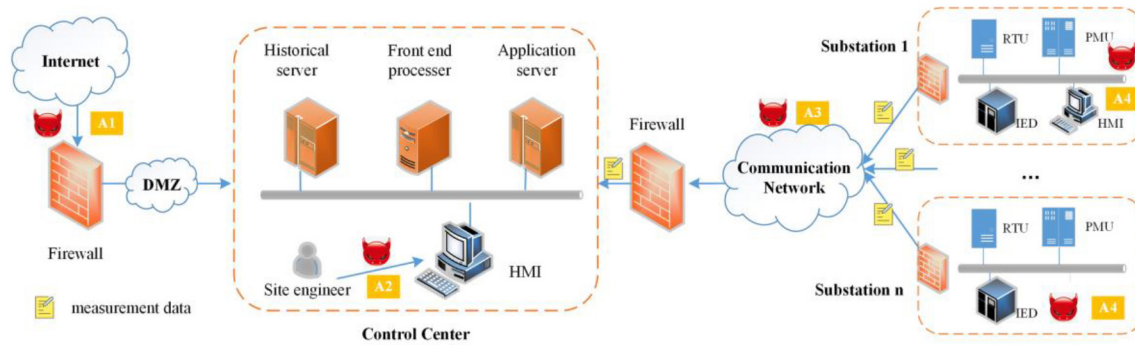


Fig. 1. Network intrusion ways in an illustrative SCADA system.

measurement data transmitted from substations and displays the information to the operator through the human-machine interface (HMI), which ensures the system awareness and controllability. The information system intrusion is the prerequisite for a cyberattack on the power system, which can be executed in the following four ways:

1) A1: Intrude the office intranet of the power grid company from the Internet, and then intrude the SCADA system. Under such circumstances, firewall and demilitarized zone (DMZ) need to be breached. But these critical cyber infrastructures usually feature sufficient defense resources, which are difficult to be intruded;

2) A2: Intrude the mobile devices of the site engineers. The attacks inject malicious code into the mobile devices of the site engineers of SCADA system, and launch the FDIA during the system upgrade or maintenance [33]–[34]. Consider that the control center features a high protection level, and the site engineers is usually fixed. Then, such intrusion targeting the site engineers' device is difficult to be executed;

3) A3: Intrude the communication network. The data transmission in the power system is mainly ensured by standard communication protocols, such as IEC101~ IEC104, IEC 61850, and IEC62351. However, these protocols are demonstrated to be with some security vulnerabilities [35]–[36], and the attacker can obtain the information needed to construct FDIA using content addressable memory table (CAM) overflow attacks, masquerading attacks, DoS attacks, etc. For instance, many messages transmitted in the substation using IEC61850 are plain text, the attacker can easily intercept the measurement data by launching a “man-in-the-middle attack” [37].

4) A4: Intrude the substation devices. The devices of a substation automation system can be organized into three levels, the station, bay, and process. Hong *et al.* [6] pointed out that the attackers could gain access to bay-level devices to launch the attacks by damaging the HMI, the station-level communication buses and so on. In addition, a large number of substations are widely distributed with various kinds of devices, e.g., phasor measurement unit (PMU), remote terminal unit (RTU), intelligent electronic device (IED). Compared with the SCADA control center, many vendors and engineers use smart devices or remote networks to maintain

substations, making it challenging to protect the confidential information in the substation. If the attackers controlled the devices or networks of these designated engineers, they can intrude the substation network and get the confidential measurement data via various means, e.g., operating system loopholes, port scanning tools, zero-day vulnerabilities [17].

On the one hand, the power system as a critical infrastructure is directly related to the safe and stable operation of the country. Thus, many countries have strengthened the protection of the critical power grid information by setting up security measures. On the other hand, the available resources for attackers are usually too limited to access the strictly confidential network information of the power system. Therefore, the traditional attack method that heavily rely on the network topology and line parameters obtained from the A1 and A2 paths is difficult to launch an effective attack in actual scenarios. Here, we focus on two viable intrusion options, i.e., A3 and A4, to construct FDIA only with measurement data in the attacking region. It is worth noting that “without network information” means the proposed attack method only need the tie line information (connection relationship and power measurement data) between the attacking region and the non-attacking region. In this paper, the problem involving FDIA and bad data detection is considered as an offensive and defensive game. Inspired by the generative adversarial network (GAN), we propose a new attack method to capture physical models and power laws contained in the measurement data via an adversarial process between the generative and adversarial networks. Also, the self-attention mechanism is utilized to determine the relationship between buses and lines. After the iterative training process, we can construct an optimal attack model that can bypass the bad data detection of the AC state estimation. The main contributions of this paper are summarized as follows.

1. A novel and feasible data-driven FDIA method is proposed. The self-attention generative adversarial network (SA-GAN) is introduced to capture the required physical information using the historical measurement data in the attacking region and tie line information between the attacking region and the non-attacking region, which could significantly reduce the demand for power system network information as compared with existing methods.

2. This method features significantly enhanced computational efficiency. After the adversarial process between the generative and adversarial networks combining the self-attention, the optimal attack model is established to directly generate false data, which avoids substantial computational complexity caused by complex iterative calculations in most traditional attack methods.

3. The performance of the proposed attack method in terms of success rate is comprehensively demonstrated under different attack intensity and three levels of measuring errors on IEEE14 and IEEE118 systems, which validates its effectiveness and scalability.

The rest of this paper is organized as follows. Section II reviews the principle of AC FDIA. Section III introduces the attack model construction and illustrates the process to construct false data. Section IV demonstrates the experimental results to verify the effectiveness of the method using IEEE14 and IEEE118 systems. Section V concludes this paper.

II. PRINCIPLES OF FALSE DATA INJECTION ATTACK

In a modern power system, various sensors are deployed to measure the real-time operation data, which will be collected by the control center. According to the weighted least squares estimation method in (1), the system state is estimated as the reference for scheduling power system dispatch.

$$\hat{\mathbf{x}} = \underset{\mathbf{x}}{\operatorname{argmin}} [\mathbf{z} - h(\mathbf{x})]^T \mathbf{R}^{-1} [\mathbf{z} - h(\mathbf{x})] \quad (1)$$

where $\mathbf{z}$ represents the measurement data (i.e., power injections at buses and flows of lines), $\mathbf{x}$ represents the system state, $\hat{\mathbf{x}}$ represents the estimated system state that fits the measurements data $\mathbf{z}$ best, and h represents the nonlinear function between measurement data and system state, which is related to the topology information and line parameters [37].

Due to possible measuring errors and disruptions, there is $\mathbf{z} \neq h(\hat{\mathbf{x}})$. Then, system operators could filter abnormal data based on bad data detection, stated as:

$$r = \|\mathbf{z} - h(\hat{\mathbf{x}})\|_2 \quad (2)$$

$$r \leq \tau \quad (3)$$

where r represents the residual when the estimated system state is $\hat{\mathbf{x}}$, and τ represents the threshold set by operators. When the residual exceeds the given threshold, it is considered that there exist abnormal data in $\mathbf{z}$.

A successful FDIA means the false data can bypass the bad data detection in the AC state estimation and pretends as normal data. Here, $\mathbf{z}$ represents a set of measurement data before the attack, and $\mathbf{a}$ represents the attack vector, where the zero position indicates that the corresponding measurement data are not attacked. The measurement data $\mathbf{z}_a$ under attack is stated as $\mathbf{z}_a = \mathbf{z} + \mathbf{a}$, and the residual r_a after the attack can be calculated as:

$$\begin{aligned} r_a &= \|\mathbf{z}_a - h(\hat{\mathbf{x}}_a)\|_2 \\ &= \|\mathbf{z}_a - h(\hat{\mathbf{x}}_a) + h(\hat{\mathbf{x}}) - h(\hat{\mathbf{x}})\|_2 \\ &= \|\mathbf{z} - h(\hat{\mathbf{x}}) + \mathbf{a} - h(\hat{\mathbf{x}}_a) + h(\hat{\mathbf{x}})\|_2 \end{aligned} \quad (4)$$

Using (2) and (3), for evading the bad data detection and launching a successful AC FDIA, the attack vector $\mathbf{a}$ should

satisfy:

$$\mathbf{a} = h(\hat{\mathbf{x}}_a) - h(\hat{\mathbf{x}}) \quad (5)$$

According to (2)-(5), the nonlinear function h between the measurement data and the power system state in AC state estimation is the core of false data construction. But the calculation of h is closely related to the network information, which is generally stored in the SCADA control center with high-level protection. Thus, the network information is usually considered to be inaccessible to potential attackers. Comparatively, the measurement data collected at a fixed time interval is considered to be much easily accessible to attackers due to the fragility of the communication network, as shown in Fig. 1.

Therefore, this paper proposes an FDIA method based on SA-GAN to launch an attack with only measurement data. The historical measurement data are fed into the network to train the attack model, where the physical model and power laws hidden in the data are learned via an adversarial process between the generator and the discriminator. The self-attention mechanism is introduced to further learn the inherent rules inside the measurement data sequence. In this way, the constructed false data can bypass the existing bad data detection algorithms and execute effective attack actions.

III. FDIA METHOD BASED ON SA-GAN

A. The Attack Principle of SA-GAN Method

The power system measurement data mainly include power injections at buses and flows of lines, and the residual is usually below the threshold of the bad data detection algorithm under normal operations. Based on these data, the operator estimates the system state via AC state estimation and executes the dispatch operation. The potential attackers aim to make the operator fail to identify the corrupted measurement data, misjudge the system state, and then result in improper dispatch scheduling, which would cause economic losses, security issues, and even system collapse.

The normal operation of the power system should obey the physical laws, such as KCL and KVL. The measurement data collected from different measurement devices at the same time-point form a snapshot of the system, which is the numerical embodiment of the physical law. These data approximately satisfy the equation constraints of power flow, and the residual after state estimation is less than the given threshold.

Fig. 2 illustrates the principle of the traditional attack method and the machine learning attack method. The traditional FDIA is model-driven, and aims (Target 1) to construct false data that obey the laws of KCL and KVL and bypass the bad data detection. The attack model f_0 relies on the network topology and line parameters to construct the false data by solving the power flow problems. Considering the difficulty of obtaining network topology and line parameters, we use a data-driven method to learn the power system physical laws contained in the measurement data with less network information and construct an attack model that can convert the normal measurement data $\mathbf{z}_1$ to the false data $\mathbf{z}_a$. It is worth noting that $\mathbf{z}_1$ and $\mathbf{z}_a$ are numerically different (to achieve

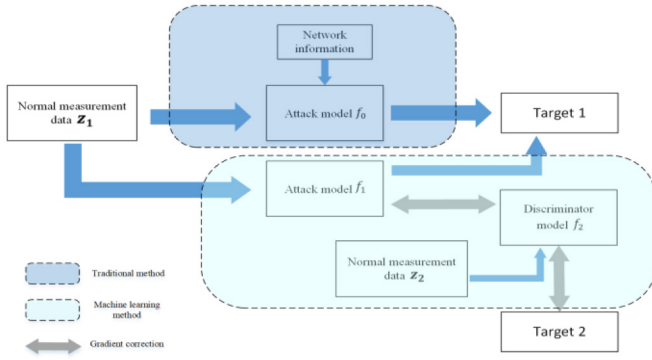


Fig. 2. The difference between the traditional attack method and the machine learning attack method.

a specific attack effect) while both of them obey the KCL and KVL laws.

Supervised machine learning is a typical data-driven method that solves the functional relationship between input data and output data. Specifically, in FDIA model f_1 , the input is the normal measurement data and the output is the false data. The premise of using the supervised machine learning method is that each normal measurement data should have corresponding false data as a label to optimize the mapping function in the attack model. However, in the model training stage, we only have normal measurement data and convert Target 1 to Target 2. Instead of using label data to optimize the model, in Target 2, we try to make the distribution of the constructed false data z_a consistent with that of the normal measurement data z_2 as much as possible. In this way, the attack model f_1 converts a set of normal measurement data into another set of “normal” measurement data, which is capable of misleading the system operation while bypassing the normal bad data detection methods.

We adopt generative adversarial network (GAN) [38] to obtain the mapping function of the attack model f_1 . Here, f_1 acts as a generator to construct false data based on normal measurement data, while f_2 acts as a discriminator to distinguish between constructed false data and normal measurement data (Target 2). The discriminator evaluates whether the false data constructed by the generator is more consistent with the laws of KCL and KVL, and supervises the generator to construct false data with higher consistency.

To better illustrate the mathematical logic of the attack model constructed by machine learning in this article, we first simplify the problem of false data injection attacks, and regard FDIA and bad data detection as the offensive and defensive game. On one hand, the attacker constructs false data z_a based on the normal measurement data z to minimize the distribution difference between z_a and z for bypassing the detection scheme deployed by the defender. On the other hand, the defender establishes a detection model to identify the false data to the greatest extent. Thus, the corresponding offensive and defensive game is formed as a minimax problem, stated as:

$$\min_{\mathcal{L}} \max_{\mathcal{R}} S(Z_a, Z) = \mathcal{L}(P_{Z_a=\Psi(Z)} || P_Z) + \mathcal{R}(Z_a, Z) \quad (6)$$

where Z_a , and Z represent the false data set and the normal measurement data set, respectively, Ψ represents the generation model that converts normal data into false data. $\mathcal{L}$ is the function to calculate the distribution difference between these two distributions, $\mathcal{R}$ is the function to distinguish between Z_a and Z . Considering the goals of the attacker and defender, the loss function (6) is to minimize $\mathcal{L}$ to construct a mapping function that can make the distribution of the constructed false data more consistent with that of the normal measurement data, and meanwhile maximize $\mathcal{R}$ so that the detection model can identify false data.

According to the aforementioned generalized representation, we proposed a SA-GAN attack method to model and solve the offensive and defensive game between FDIA and bad data detection through an adversarial process.

B. Attack Model and Training Method

The proposed FDIA model based on SA-GAN mainly includes three modules: generator module, discriminator module, and self-attention module, as depicted in Fig. 3.

1) *Generator Module and Discriminator Module:* For a given time-point, the composition of the measurement data z is stated as:

$$z = \{P_i, Q_i, P_j, Q_j, p_{ij}, q_{ij}\}, i, j \in S_A \quad (7)$$

where S_A represents the set of buses in the attacking region, P_i, Q_i, P_j , and Q_j represent the power injections at bus i and j , respectively, p_{ij} and q_{ij} represent the power flows of lines connecting bus i and j . To ensure the false data constructed by the generator different from the historical measurement data, the training data set Z composed of many measurement data samples is divided into data set Z_1 and data set Z_2 , where $z_1 \in Z_1$ are as input to the generator module, and $z_2 \in Z_2$ are as input to the discriminator module.

As discussed in Section III-A, the minimax problem in FDIA can be solved by the two-player minimax game between the generator module and the discriminator module. The generator module G aims to construct false data with a similar distribution as the normal data for bypassing the bad data detection. Therefore, the generator receives data z_1 to be attacked as input, constructs a mapping function $G(z_1, \theta_g)$ from the distribution $\mathcal{F}_{Z_1}$ of z_1 to the normal data distribution $\mathcal{F}_{Z_2}$ (θ_g is the mapping parameter to z_1), and then generates a group of false data $G(z_1)$. To determine whether the received measurement data have been tampered with, the discriminator try to find a hyperplane to separate z_2 from $G(z_1)$ and output the probabilities that z_2 and $G(z_1)$ are belong to the normal measurement data.

During the training process, in order to avoid overfitting we alternate between k steps of optimizing D (stage one) and one step of optimizing G (stage two). At stage one, the generator parameters (the mapping function $G(z_1, \theta_g)$) are maintained unchanged, and the discriminator finds a hyperplane based on the constructed false data $G(z_1)$ and the measurement data z_2 to classify these two types of data. Here, it is easy for the discriminator to find hyperplanes to classify them due to the randomness of the initialized generator parameters and the low

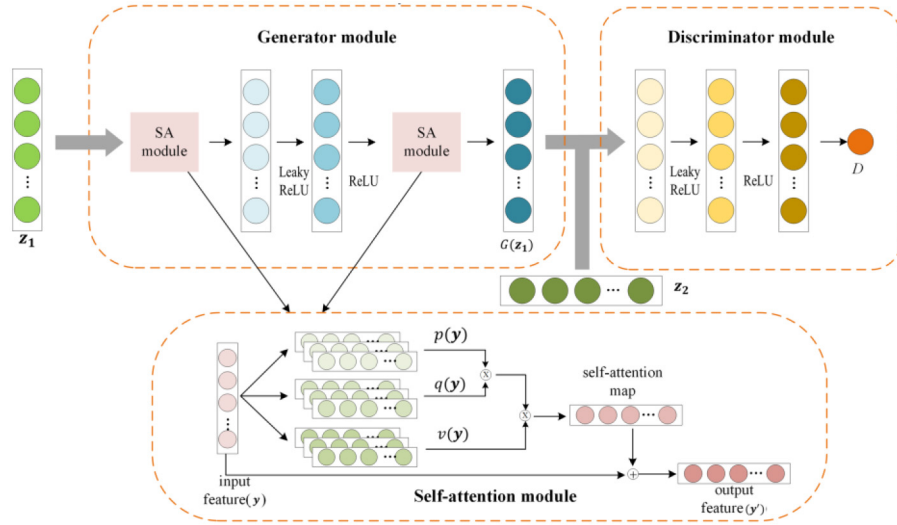


Fig. 3. Attack model based on SA-GAN.

fidelity of the constructed false data. In stage two, the discriminator parameters are maintained unchanged, and the generator parameters will be optimized. The generator calculates the distance between the false data $G(z_1)$ and the divided hyperplane, and minimizes the distance by optimizing the generator parameters to cheat the discriminator module. Through multiple adversarial processes, the network parameters of G and D are optimized iteratively. After that, the distribution of the false data $G(z_1)$ constructed by the generator will have a similar distribution as the normal measurement data z_2 and becomes difficult to be identified by bad data detection. The objective function of this adversarial process takes the minimax form as (8), which is composed of two parts, i.e., the loss function of generator $E_{z_1 \sim \mathcal{F}_{Z_1}}[1 - \log D(G(z_1))]$ and the loss function of discriminator $E_{z_2 \sim \mathcal{F}_{Z_2}}[\log D(z_2)]$.

$$\min_G \max_D V = E_{z_2 \sim \mathcal{F}_{Z_2}}[\log D(z_2)] + E_{z_1 \sim \mathcal{F}_{Z_1}}[1 - \log D(G(z_1))] \quad (8)$$

where $E(\varphi)$ represents the calculation of the expected value. z_2 represents the normal measurement data, and z_1 represents the data to be attacked. $G(z_1)$ represents the false data constructed by the generator. $\mathcal{F}_{Z_2}$ represents the distribution of normal measurement data z_2 which is as input of the discriminator, and $\mathcal{F}_{Z_1}$ represents the distribution of z_1 which is as input of the generator.

2) *Self-Attention Module*: In traditional GAN, the neurons in each layer of the generator will treat the input data from the previous layer equally without considering the correlation among these data. However, there are inherent couplings between buses and lines in the power system. More specifically, power injections at different buses and the flows of different lines would have different impacts on power system operations according to KCL and KVL laws. Thus, it is essential to re-organize the GAN structure to exploit the inherent coupling in the measurement data sequence during the adversarial process.

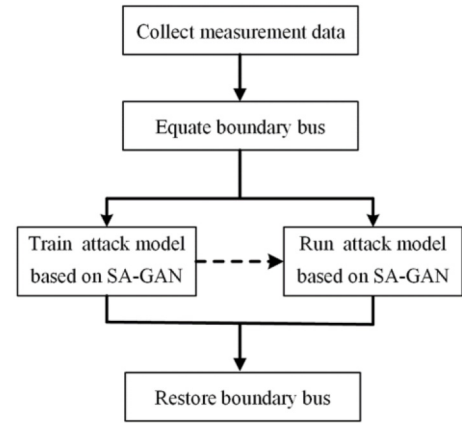


Fig. 4. Construction process of false data.

The self-attention mechanism [39] has been demonstrated as a viable solution for extracting the constraint relationship between elements at different positions in the data sequence, focusing on the interactive representation between sequence elements, and capturing the internal correlation of data or features. It is validated in various tasks of distinct fields [40]–[42]. In [43], the self-attention module is integrated into the generative adversarial network to redistribute the attention weight between data sequence elements. In this way, the GAN can learn the correlation in the data sequence and then ensure better consistency between the power injections at buses and the flows of lines in the constructed false data. Like other GAN networks, SA-GAN faces computational complexity and instability issues. In this paper, we adopt spectral normalization and activation function in the training process to cope with these potential issues [41].

C. False Data Construction Process

Fig. 4 depicts the construction process of false data, and the detailed procedures are stated as follows.

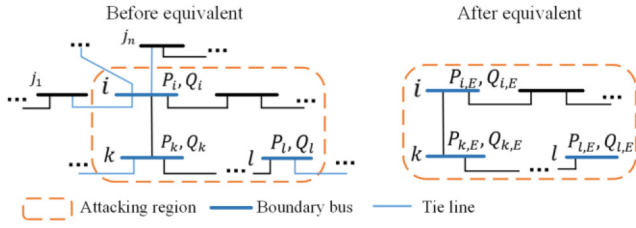


Fig. 5. Calculation of equivalent injection power at boundary buses.

1) Collect measurement data. To ensure the authenticity and regularity of the data, data set Z is constructed based on the measurement data collected by the SCADA of a power system, and the composition of the measurement data is stated in (7).

2) Equate boundary bus. In this paper, the power system can be divided into attacking and non-attacking region. In Fig. 5, some lines connected to the boundary bus i belong to the non-attacking region, according to the KCL and KVL formulas, the active power injection P_i and reactive power injection Q_i of bus i satisfies (9)-(10), and the power of the boundary bus i after equivalent can be calculated as (11)-(12):

$$P_i = \sum_{j \in S_{i,N}} p_{ij} + \sum_{j \in S_{i,A}} p_{ij} \quad (9)$$

$$Q_i = \sum_{j \in S_{i,N}} q_{ij} + \sum_{j \in S_{i,A}} q_{ij} \quad (10)$$

$$P_{i,E} = P_i - \sum_{j \in S_{i,N}} p_{ij} \quad (11)$$

$$Q_{i,E} = Q_i - \sum_{j \in S_{i,N}} q_{ij} \quad (12)$$

where $S_{i,N}$ is the set of buses in the non-attacking region connected to bus i , $S_{i,A}$ is the set of buses in the attacking region connected to bus i . p_{ij} and q_{ij} represent the active and reactive flow of the line with bus i as the starting point and j as the end point, respectively, $P_{i,E}$ and $Q_{i,E}$ are the active and reactive power injected into the bus i after equivalent.

3) Train attack model based on SA-GAN. According to the general division ratio of the data set in machine learning, the historical measurement data set Z is divided into training set T_1 , validation set T_2 and test set T_3 at a ratio of 4:2:4. To ensure the objectivity of the evaluation of the attack model, there should be no intersection between the training set T_1 , the validation set T_2 and the test set T_3 . Therefore, they need to satisfy the following constraints:

$$T_1 \cup T_2 \cup T_3 = Z \quad (13)$$

$$T_1 \cap T_2 = \emptyset, T_1 \cap T_3 = \emptyset, T_2 \cap T_3 = \emptyset \quad (14)$$

Fig. 6 depicts the relationship between the residual caused by the false data and the number of iterations during the training process. As the adversarial process between the generator and discriminator continues, the residual of the false data constructed by the attack model gradually decreases, which indicates that the constructed false data would be more and more difficult to be identified by the bad data detection algorithms.

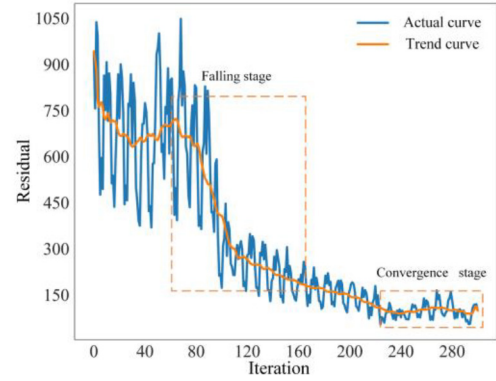


Fig. 6. The relationship between the residual and iterations.

4) Run the attack model based on SA-GAN. Given the obtained optimal attack model parameters, the measurement data in the test set T_3 are fed into the attack model to generate the false data of the buses and the lines in the attacking region.

5) Restore boundary bus. In Step 4), the tampered data of the buses and lines in the attacking region are obtained, while the flow of tie line connected to boundary bus i is not included in the tampered data $P'_{i,E}$ and $Q'_{i,E}$. With a reverse process of Step 2), and the final tampered injection power P'_i and Q'_i of the boundary bus i in the attacking region can be obtained as:

$$P'_i = P'_{i,E} + \sum_{j \in S_{i,N}} p_{ij} \quad (15)$$

$$Q'_i = Q'_{i,E} + \sum_{j \in S_{i,N}} q_{ij} \quad (16)$$

IV. CASE STUDIES

In this section, the effectiveness of the proposed attack method is validated using two test systems, i.e., the IEEE14 system and the IEEE118 system provided by Matpower7.0. In the experiment, the network information in the standard system is kept unchanged. Each bus and each line is equipped with a power measurement to get active and reactive power. Therefore, for the snapshot at a given time-point, the number of measurement data of the IEEE14 and IEEE118 systems are 68 and 594, respectively.

Here, the operation data of some substations located in a certain region of eastern China are sampled every 15 minutes and then construct the sample dataset after data masking. To simulate the measuring error of the equipment, $N \sim (0, 1)$ standard normal noise is added into all measurement data to simulate the real power flow data. We construct 20,000 pieces of sample data, of which 8,000 pieces are used as the training set T_1 for the learning of network parameters of the attack model, and 4000 pieces are used as the verification set T_2 to test the training effect of the model. The remaining 8000 pieces are used as test set T_3 to evaluate the performance of the optimal attack model on new data.

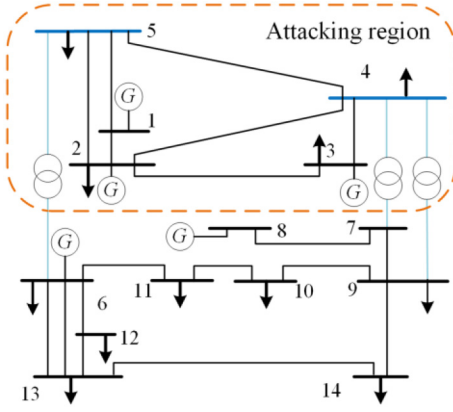


Fig. 7. Region division of IEEE14 system.

TABLE I
BUSES POWER BEFORE AND AFTER THE ATTACK

Bus	Before attack		After attack	
	P (MW)	Q (MVar)	P' (MW)	Q' (MVar)
1	345.2011	-18.8337	188.2251	-1.9671
2	8.5247	95.1034	8.8739	42.8177
3	-124.5949	42.5698	-66.4602	8.8351
4	-67.6323	-46.7257	-23.3347	-28.3727
5	-10.3692	-7.2176	29.9514	-18.0377
6	-16.3438	91.8102	-16.2804	91.9369
7	1.1625	-1.0947	1.5892	-1.2058
8	-1.3855	51.3327	-0.9592	51.2383
9	-47.2164	-32.5799	-46.8786	-32.6789
10	-13.0853	-8.6906	-12.8	-8.7484
11	-3.3849	-1.0059	-3.2135	-0.9707
12	-6.086	-6.1934	-5.9973	-6.072
13	-13.6559	-10.3937	-13.5516	-10.2902
14	-26.9679	-20.1213	-26.7256	-20.1225

A. Construction of False Data

Fig. 7 depicts the division of attacking and non-attacking regions of the IEEE14 system. The attacking region includes buses 1, 2, 3, 4, 5 and lines 1-2, 1-5, 2-3, 2-4, 2-5, 3-4, 4-5. Lines 4-7, 4-9, and 5-6 are the tie lines between the attacking and non-attacking region.

For the IEEE14 system, the specific parameters of the attack model during training are set as follows: we operate 300 iterations and gather 200 pieces of sample data in each batch of the training set, and each piece of sample data contains 24 measurement data. In addition, we choose RMSprop as the optimizer, and set the learning rate as 0.0003 and the moving average coefficient α as 0.9. In Fig. 7, there are 24 and 44 measurement data in the attacking and non-attacking regions, respectively. Here, supposing the attacker tampers with the measurement data at one time-point, only the 24 measurement data in this piece of sample data need be falsified.

Table I presents the buses power before and after the attack, and Table II presents the power measurement data of the three boundary tie line before the attack. Based on (11)-(12), the injection powers of the boundary buses 4 and 5 in the attacking region are equated. After equivalent processing, the active

TABLE II
MEASUREMENT DATA OF THE TIE LINE BEFORE THE ATTACK

Line	From	To	p (MW)	q (MVar)
8	4	7	39.2782	-10.2912
9	4	9	21.9186	3.0164
10	5	6	64.9578	-23.8706

and reactive power measurement data of boundary bus 4 in the attacking region are -128.8291 MW and -39.4509 MVar, respectively, and the active and reactive power measurement data of bus 5 are -75.327 MW and 16.653 MVar, respectively.

The attacker directly inputs the equivalent 24 measurement data into the trained SA-GAN model to construct false data, and then restore the constructed false data according to (15)-(16) to obtain the final false data.

B. Analysis of Attack Intensity

The attacker aims to hide the attack by evading the bad data detection algorithm, which can be easily realized when there is a small deviation between the false data and the normal data. However, operators rarely execute wrong operations based on these attacked data with a small modification. In this paper, the attack intensity γ of false data constructed by our proposed method is calculated when the measuring error follows the standard normal distribution.

$$\gamma = \frac{\sum_{i \in S_A} (|P_i - P'_i| + |Q_i - Q'_i|) + \sum_{j,k \in S_A} (|p_{j,k} - p'_{j,k}| + |q_{j,k} - q'_{j,k}|)}{\sum_{i \in S_A} (|P_i| + |Q_i|) + \sum_{j,k \in S_A} (|p_{j,k}| + |q_{j,k}|)} \quad (17)$$

To evaluate the attack intensity of the false data on the original measurement data, the ratio of the absolute difference (the absolute difference between the data after the attack and the data before the attack) to the data before the attack is defined as the attack intensity and calculated according to (17). S_A is the set of buses in the attacking region, i, j and k are the subscripts of the buses in the attacking region, and $P_i, Q_i, p_{j,k}, q_{j,k}$ and $P'_i, Q'_i, p'_{j,k}, q'_{j,k}$ are the power data of the buses and lines in the attacking region before and after the attack, respectively. Fig. 8 shows the attack intensity distribution of the 8000 pieces of sample data in the test set attacked by the SA-GAN model, where 7959 pieces of injected false data evade the bad data detection with an average residual of 3.877. It can be seen that the attack intensity mostly ranges between 42% and 48%, peaks around 53.39%, and averages at 45.17%. In Table I, there exist significant deviations between the bus power before and after the attack at buses 1, 2, 3, 4, and 5 in the attacking region. Specifically, the active power deviation at bus 1 reaches 156.976 MW, and the corresponding attack intensity is 45%. It is demonstrated that there would exist large deviations between the injected false data and the normal measurement data, and these injected false data would mislead the system operators and deteriorate the power system operations.

C. Analysis of Attack Success Rate

For the SCADA system, the measuring error often leads to deviations between the original measurement data and the

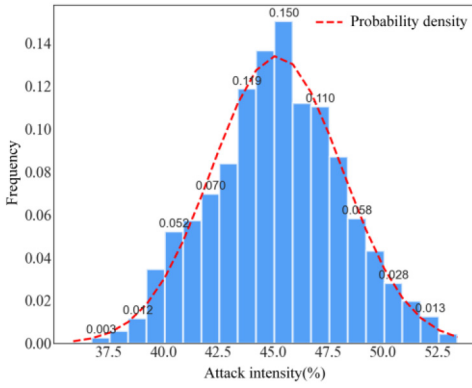


Fig. 8. Attack intensity distribution under standard normal error.

TABLE III
THE SUCCESS RATE OF IEEE14 SYSTEM UNDER DIFFERENT ERRORS

Normal noise	Residual before attack	Residual after attack	Success rate
	min/mean/max	min/mean/max	
$N \sim (0,1)$	2.964/16.059/47.812	0.159/3.877/18.928	99.48%
$N \sim (0,1.15)$	3.426/19.299/64.519	0.234/4.363/29.948	99.71%
$N \sim (0,1.25)$	4.554/25.013/80.089	0.178/5.104/30.261	99.78%

power flow data after state estimation, which may affect attacks. Therefore, three normal noises $N \sim (0, 1)$, $N \sim (0, 1.15)$, and $N \sim (0, 1.25)$ are added to the sample data for simulating measuring error and the attack success rate of the proposed SA-GAN based attack model is calculated and analyzed.

By analyzing the attack principle of III-A, the SA-GAN attack model proposed continuously optimizes the parameters of the modules using game on the entire historical measurement data to make the constructed false data obey the KCL and KVL laws as much as possible. Here, we calculate the residual and success rate after the attack under different measuring errors, and the results are shown in Table III. The judgment criterion of a successful attack is that the residual of the power system after the attack is less than that before the attack.

As the measuring errors increase, the residual of the false data constructed by the proposed SA-GAN based attack model is always smaller than the residual before the attack. Besides, when the variance of the measuring error reaches 1.15 and 1.25, the average residual of the system before the attack increase by 3.234 and 8.954, while those after the attack only increase by 0.486 and 1.227. It is demonstrated that the false data constructed by the proposed SA-GAN based attack model is capable of evading the bad data detection regardless of large measuring errors. In other words, the proposed attack model features good robustness against measuring errors.

D. Analysis of Time Efficiency

Apart from the effectiveness of the false data, the attack should construct the false data and inject them into the power system in a timely fashion. Most existing cyberattacks targeting AC state estimation are based on the iterative calculation of the Newton-Raphson method. Their time efficiency is relatively low, resulting in that they might be incapable of

construct effective false data timely, especially for large-scale power systems. Here, the time taken to solve the system power flow equation by the Newton-Raphson method is considered as the approximately required time of these traditional iterative attack methods, which will be compared with the consumed time for constructing false data using the proposed SA-GAN based attack model.

In the IEEE14 system, the proposed SA-GAN based attack model averagely takes 0.57ms to construct an effective FDIA, and the traditional iterative attack method averagely requires about 17.41ms to finish the same task. In terms of time efficiency, the proposed SA-GAN based attack model works nearly 30 times faster than traditional iterative attack methods. Compared with these traditional iterative attack methods, the proposed SA-GAN based attack model eliminates the need for a complicated iterative process and features significantly enhanced time efficiency.

E. IEEE118 System Performance Test

Here, the attacking region of the IEEE118 system consists of 10 buses (i.e., buses 103-112) and 15 lines (i.e., lines 156-170). Among them, lines 156, 157, and 160 are the tie lines between the attacking and the non-attacking region. The beginning and end buses of the three tie lines are 100-103, 100-104, and 100-106, respectively. The data equivalent process is similar to that of the IEEE14 system in Section IV-A.

For the IEEE118 system, the parameters of the proposed SA-GAN based attack method during training are set as follows: we operate 300 iterations and gather 200 pieces of sample data in each batch of the training set and test set, and each piece of sample data contains 44 measurement data. Then, we choose RMSprop as the optimizer, and set the learning rate as 0.00003 and the moving average coefficient α as 0.7.

Adopting the same judgment criterion of successful attack in Section IV-C, we analyze the data satisfying that the residual of the power system after the attack is less than that before the attack. The statistical results are shown in Table IV, in which the measuring error follows a standard normal distribution, and the average residuals before and after the attack are 35.0988 and 15.5358 respectively. Moreover, the maximum and minimum residuals after the attack are always smaller than those before the attack. This indicates that the constructed false data would be difficult to be identified by the bad data detection. Also, the success rate of the constructed FDIA reaches 93% in the tested IEEE118 system, which validates the scalability of the proposed SA-GAN based attack method. The consumed time of the proposed SA-GAN based attack method and traditional iterative attack methods are 3.47ms and 40.47ms, respectively. The proposed attack method works nearly 11 times faster than these traditional methods. Then, the superiority of the proposed SA-GAN based attack method over the traditional iterative attack methods in terms of time efficiency is demonstrated. Accordingly, the proposed SA-GAN based attack method provides a viable solution for constructing effective FDIA in a timely fashion for large-scale power systems.

TABLE IV
INDEXES UNDER IEEE118 SYSTEM

Index	Value	
	Before the attack	After the attack
Maximum residual	100.5110	53.7974
Mean residual	35.0988	15.5358
Minimum residual	11.3633	4.0673
Success rate	93%	
Time efficiency	3.47ms	

V. CONCLUSION

The massive adoption of ICTs infrastructure renders power systems not only increase the system efficiency but also introduce potential cyberattack risks. The development of effective cybersecurity countermeasures relies on a deep understanding of potential cyberattacks. In this paper, we propose a novel SA-GAN based FDIA method for constructing effective FDIA promptly with less system network information. In this attack method, FDIA construction is modeled as a minimax problem, and the generative adversarial network integrated with a self-attention mechanism is adopted to extract the physical model and power laws hidden in historical data. Compared with existing attack methods, the proposed data-driven attack model can reduce the demand for power system network information and eliminates complicated iterative processes, featuring significantly enhanced feasibility and time efficiency.

Extensive case studies have been conducted in the IEEE14 and IEEE118 systems, where the success rate of the proposed SA-GAN based FDIA model is always maintained above 90%. Also, it is demonstrated that the proposed FDIA model presents desired robustness against measuring errors and high time efficiency, which ensures the effectiveness of the proposed method under varying system operating conditions. Accordingly, this paper provides a comprehensive guide to researchers and practitioners interested in potential cyberattacks of power systems for developing advanced countermeasures and enhancing cybersecurity capabilities of modern power systems.

In the future, we will continue to explore the following issues.

(1) How to train the model when the power system changes. In response to this problem, we plan to consider adding dynamic constraints to simulate the impact of network topology and line parameters changes when constructing the data and training attack models.

(2) How to develop effective detection and mitigation algorithms for this type of attack. Based on the analyses of the attack principle, we demonstrate the prominent characteristics of the attack and the necessary conditions for launching an effective attack. This could guide us to design targeted detection and defense algorithms to effectively cope with such attacks and maintain the safe and stable operation of the power grid.

(3) How to guarantee the robustness of the attack model. Cybersecurity involves complex human behaviors, political science, organizational behavior, and other factors. We have conducted preliminary considerations on some relevant factors

when constructing the proposed attack model. In the future, we will include more reality factors to construct an attack model with high robustness and feasibility in practical fields.

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